

Seawulf MPI: Quick Instructions

Quick reference for allocating resources, loading MPICH, compiling, running, and logging the ordered MPI program.

1. Allocate 16 MPI tasks

```
salloc --partition=short-28core --nodes=1 --ntasks=16 --cpus-per-task=1
```

2. Enter the allocated compute node

```
srun --pty bash
```

3. Load MPICH and GCC

```
module load mpich/gcc/3.4.2
```

Loading this module also loads the compatible GCC compiler and required dependencies.

4. Compile the ordered C program

```
mpicc hello_ordered.c -o hello_ordered
```

mpicc is the MPI C compiler wrapper. It invokes GCC with the required MPI headers and libraries.

5. Run the ordered program

```
srun --mpi=pmi2 -n 16 ./hello_ordered
```

This launches 16 MPI processes. On this Seawulf setup, `--mpi=pmi2` allows the Slurm tasks to participate in the same MPI job.

6. Create `ordered_log.txt` with three runs

Use `>` for the first heading to create/reset the file, then `>>` to append every subsequent heading and run.

```
echo "Run 1:" > ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt

echo "Run 2:" >> ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt

echo "Run 3:" >> ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt
```

Complete Workflow

```
salloc --partition=short-28core --nodes=1 --ntasks=16 --cpus-per-task=1
srun --pty bash
module load mpich/gcc/3.4.2

mpicc hello_ordered.c -o hello_ordered

echo "Run 1:" > ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt

echo "Run 2:" >> ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt

echo "Run 3:" >> ordered_log.txt
srun --mpi=pmi2 -n 16 ./hello_ordered >> ordered_log.txt
```